10.5 Angle Relationships in Circles

Name _____ Period ____ Date _____

Geometry 10.5 (30-minute plan) · Student sheet

I can…

- find the angle formed by a tangent and a chord.
- find angles formed by lines that meet inside or outside a circle.
- find the measure of a circumscribed angle.

Vocabulary · fill in the missing word

Circumscribed angle — an angle whose sides are _____ to a circle

Secant — a line that _____ a circle in two points

Tangent — a line in the plane of a circle that _____ the circle in exactly one point

Part 1 · Notes and examples (fill in each note, then work the example)

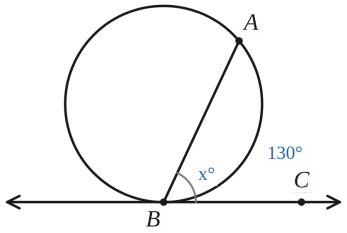
Tangent and Intersected Chord Theorem: the angle formed by a tangent and a chord at the point of tangency is _____ its intercepted arc.

Example 1

Line BC is tangent to the circle at B.

a. Find m∠ABC.

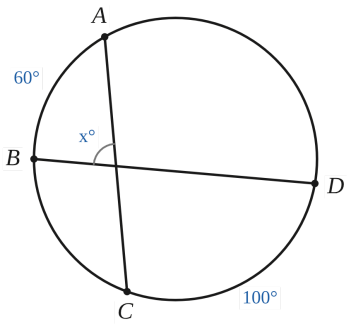
b. If the angle formed by a tangent and a chord is 104°, what is its intercepted arc?



Angles Inside the Circle Theorem: when two chords cross inside a circle, each angle is one-half the _____ of the arcs intercepted by the angle and its vertical angle.

Example 2

Two chords cross inside the circle. Find the value of x.

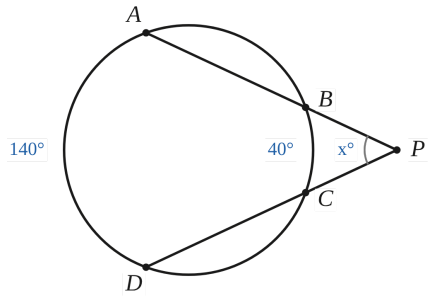


Angles Outside the Circle Theorem: when tangents or secants meet outside a circle, the angle is one-half the _____ of the intercepted arcs (far arc minus near arc).

Circumscribed Angle Theorem: a circumscribed angle measures 180° minus the _____ angle that intercepts the same arc.

Example 3

Two secants meet at P outside the circle. Find the value of x.



Part 2 · Partner practice

Partner practice

A tangent and a secant meet outside a circle. The far arc measures 160° and the near arc measures 50°. Find the angle.